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Roll No

ME-801(C)-GS

B.E. VIII Semester

Examination, December 2020

Grading System (GS)

Reliability and Maintenance

(Elective - III)

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Discuss evaluation of TPM. Explain FMEA.
b) Explain about FMECA. Write its elements, applications and benefits.
2. a) What do you understand by maintenance planning and scheduling? Explain its significance.
b) Compare centralised and decentralised maintenance organisation. State the advantages of each one over another.
3. Discuss the features that constitute TPM. What steps and procedures should be taken up to adopt TPM in existing industries in India?
4. Explain Weibull distribution. How reliability of system can be computed with this distribution.

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PTO

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5. A pumping station has two identical pumps connected in parallel, each capable of supplying 3000 lit/hr. the failure rate and repair rate of each is 0.5 f/hr and 4 r/hr respectively. Evaluate the frequency of encountering the duration of residing in each possible throughput state.
6.
 - a) Differentiate between preventive and corrective line the level of preventive maintenance that may be needed eel keeping overall cost of maintenance in mind.
 - b) What is design out maintenance? Is it possible to achieve the objectives of design out maintenance in all operating system. Briefly explain?
7.
 - a) How reliability and quality are associated? Discuss.
 - b) Explain Markov analysis for reliability.
8. Write short technical notes on following (any two):
 - a) Visual monitoring
 - b) MTTR
 - c) K-out-of-m Systems
 - d) Risk Evaluation and Priorities
